Class Bias Correction Matters: A Class-Incremental Learning Framework for Remote Sensing Scene Classification

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***Abstract*— Most existing deep learning models for remote sensing scene classification (RSSC) adopt offline learning paradigm, which are trained on closed datasets and fail to dynamically update with new class data. Currently, class-incremental learning (CIL) allows models to learn new classes while retaining discrimination of old ones. However, most CIL approaches aim to overcome catastrophic forgetting by employing techniques such as exemplarmemory and knowledge distillation, while ignoring the prediction bias caused by imbalanced datasets, where old classes retain fewer samples than new ones. Moreover, they do not adequately account for the multilevel semantic structure and multiscale feature information inherent in remote sensing images (RSIs). To address these issues, we propose an effective CIL framework for RSSC, named class bias correction network (CBC-Net). Specifically, a cross-dimensional and interaction-aware attention mechanism (CIAM) is designed to incorporate channel, position, and direction-aware information in feature maps, enabling the model to highlight informative regions within RSIs. Next, a contextual information fusion module (CIFM) is proposed to explore the correlations among multilevel features and enhance representation quality through their fusion. In addition, the designed taskwise classifier head decoupling mechanism (TCDM) imposes a constraint to mitigate the prediction bias toward new classes, and enhances model's discrimination among all seen classes. Finally, a multilevel integrated knowledge distillation module (MKDM) is developed to ensure comprehensive knowledge transfer, empowering the model to maintain critical representations in feature space and make well-informed decisions in output probability space. Experiments on five open datasets demonstrate the outperformance and robustness of our method.**

***Index Terms*— Bias correction, catastrophic forgetting, incremental learning, remote sensing scene classification (RSSC).**

I. Introduction

REMOTE sensing scene classification (RSSC) is a crucial technique for assigning scene-aware labels for remote sensing images (RSIs) based on the target properties and semantic information. Recent advancement in sensor and Earth observation technologies has greatly increased the availability of RSIs, which involve diverse scenarios, requiring that the RSSC system continuously adapts to new classes.

The field of RSSC has witnessed a significant evolution in methodology [1], transitioning from handcrafted feature-based methods [2], [3], [4], [5], [6], [7] to unsupervised learning methods [8], [9], [10], [11], and currently advancing to supervised learning methods [12], [13], [14], [15], [16], [17], [18]. However, the limitations of these methods are becoming increasingly apparent in the dynamic and real-world scenarios. Most methods are offline learning algorithms, which means models are trained on closed datasets and fail to update with new class data. When a model adjusts its parameters to adapt to new classes, it may encounter a severe decline in its ability to generalize to old data, a phenomenon known as catastrophic forgetting. Besides, this paradigm proves inefficient for dynamic learning tasks as it requires retraining on all the old and new data, which consumes significant storage memory and computational resources. This inadequacy highlights the critical necessity for incremental learning algorithms.

From a biological perspective, humans accumulate knowledge incrementally, learning and adapting as new experiences present. This process inspires the incremental learning algorithms, which are designed to continuously update and refine their models with only new data while retaining the ability to accurately distinguish old classes. Meanwhile, we expect that the algorithms possess both plasticity and stability, reflecting the trade-off between retaining old knowledge and adapting to new information. Incremental learning encompasses three settings, each posing distinct challenges for machine learning models [19]. Domain-incremental learning (DIL) [20], [21], [22] focuses on the challenge that training data grows dynamically but the set of classes remains constant. Task-incremental learning (TIL) [23], [24], [25] employs a shared feature extractor, with each new task adding a classification layer. Inference in TIL requires the task ID of the test samples in advance. In contrast, class-incremental learning (CIL) [26] involves all classes sharing a classification layer that expands as new classes are learned. Inference is performed across all classes without the need to specify the task ID of samples. This makes CIL more challenging and closer to practical application scenarios.

Recently, CIL in RSSC has garnered increasing attention and some works are proposed in applications [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], which are effective in

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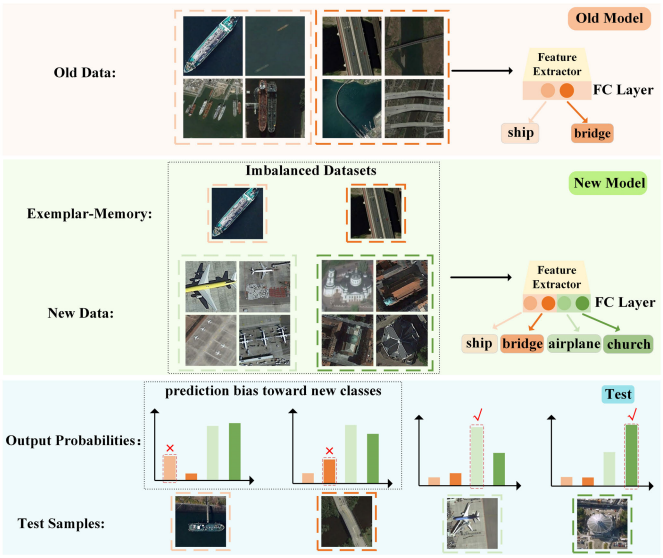


Fig. 1. Illustration of imbalanced datasets and prediction bias in CIL. The old model, trained on old classes (e.g., ship and bridge), is updated with new classes (e.g., airplane and church). Due to limited memory, the exemplar memory retains only a few samples of old classes, leading to the imbalanced amount of old and new data. This imbalance results in prediction bias toward new classes, as the model assigns higher probabilities to new classes during testing. Meanwhile, CIL in RSSC faces unique challenges because of the intricate background (e.g., church and bridge) and huge target scale variance (e.g., ship and airplane) inherent in RSIs.

mitigating the issue of catastrophic forgetting to some extent. However, they continue to face the following challenges. First, it is common and effective to retain a few old class samples in the exemplar memory (e.g., ship and bridge) among these methods, which is shown in Fig. 1. However, the replay-based methods have a significant bias toward the new classes (e.g., airplane and church) due to the imbalanced amount of old and new data, which is manifested in the higher logit values for new classes when compared to old ones. Few works have been proposed to thoroughly analyze the core causes of such bias and address this issue. Second, most existing methods do not adequately account for the multilevel semantic structure and multiscale feature information inherent in RSIs. As illustrated in Fig. 1, CIL in RSSC faces unique challenges because of the intricate background (e.g., church and bridge), huge target scale variance (e.g., ship and airplane), and multiple target classes in RSIs, requiring advanced techniques to reduce intraclass ambiguity and boost interclass distinctiveness for accurate scene discrimination. Meanwhile, standard knowledge distillation (SKD) has been naively adopted by most methods to retain knowledge from past tasks without adequate justification. This practice introduces a significant bias in the soft targets, skewing them toward the recent old tasks. Furthermore, most of these methods that employ distillation loss for RSSC typically focus solely on constraining the output probabilities for past tasks, while neglecting high-plasticity knowledge distillation (HPD) in feature representation space. This oversight results in the model's failure to maintain critical representations of old classes within feature space, thus leading to catastrophic forgetting.

To address the above issues, we propose an effective CIL framework for RSSC, named class bias correction network (CBCNet). To correct the prediction bias introduced by imbalanced datasets, the designed taskwise classifier head decoupling mechanism (TCDM) imposes a constraint to mitigate the overpenalization of the classification scores for old classes, and optimizes the model's classification capabilities for all seen classes. Meanwhile, an adaptive gradient-balanced cross-entropy (CE) loss is introduced to compensate for the imbalanced gradient contributions to the classification scores. In addition, most methods typically focus solely on constraining the output probabilities for past tasks. To counteract this constraint, a multilevel integrated knowledge distillation module (MKDM) is designed to ensure comprehensive knowledge transfer by combining the taskwise knowledge distillation (TWD) in output probability space and HPD in feature representation space. Furthermore, two novel components, cross-dimensional and interaction-aware attention mechanism (CIAM) and contextual information fusion module (CIFM), are proposed to highlight informative regions across multiple scales and aggregate contextual information from multiple levels within RSIs, thereby achieving a comprehensive understanding of the RS scenes.

The main contributions of our work are summarized as follows.

1) We systematically analyze the root causes of prediction bias toward new classes in CIL methods and then propose a CIL framework for RCCS, named CBCNet. This framework aims to effectively combat catastrophic forgetting and balance the stability–plasticity dilemma.
2) Two novel components, CIAM and CIFM, collaborate to highlight informative regions across multiple scales and aggregate contextual information from multiple levels in RSIs, thus constructing a richer feature representation.
3) The designed TCDM imposes a constraint to mitigate the prediction bias toward new classes, and enhances model's discrimination among all seen classes. Meanwhile, an adaptive gradient-balanced CE loss is introduced to compensate for the imbalanced gradient contributions to the classification scores.
4) A novel MKDM is specifically designed to ensure comprehensive knowledge transfer, empowering the model to maintain critical representations in the feature space and make well-informed decisions in the output probability space. Experiments on five open datasets demonstrate the outperformance and robustness of our method.

II. Related Work

In this section, we briefly introduce the existing research on incremental learning methods and discuss pertinent strategies that have been developed to mitigate catastrophic forgetting. Subsequently, we provide a detailed overview of the existing CIL methods for RSSC.

A. Incremental Learning

1) Regularization-Based Methods: For incremental learning, regularization strategies have become a key approach. These methods maintain the integrity of previously acquired knowledge by adding penalties to the loss function, which prevent sudden changes in model's parameters when confronted

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with new data. This approach can be generally classified into two main types: methods based on prior information and methods utilizing knowledge distillation techniques.

The methods based on prior information employ two techniques: estimating parameter importance and using subspace projection. The former provides a means to selectively regularize the learning process by identifying and protecting critical parameters. Elastic weight consolidation (EWC) [22] was the first to use parameter regularization to prevent catastrophic forgetting. It utilizes the Fisher information matrix to evaluate the importance of weights and then decelerates learning for those critical to earlier tasks. Synaptic intelligence (SI) [37] introduced a method by gauging the importance of parameters by their effect on the loss decrease and using the path integral in the optimization path. Pan et al. [38] proposed a functional regularization method that employs a Gaussian process to identify key past data points and incorporates a functional prior information. Memory aware synapses (MAS) [39] introduced a method for gauging parameter importance by analyzing the network's output gradients. Another notable technique is subspace projection, which projects the gradients from new tasks onto a subspace where outputs of old tasks remain unchanged. Gradient episodic memory (GEM) [40] proposed to impose a constraint on the direction of the parameter update vector and the gradients of the network's parameters associated with old classes.

Methods utilizing knowledge distillation technique leverage the consistency between old and new models through maintaining similar outputs, preserving feature representations, or ensuring the stability of sample relationships across different incremental steps. Learning without forgetting (LwF) [41] incorporates a distillation loss into the overall loss function to prevent catastrophic forgetting. Incremental classifier and representation learning (iCaRL) [26] stores samples for each class that are closest to the average vector and employs nearest mean of exemplars for classification. Topology-preserving class-incremental learning (TPCIL) [42] constructed the neighborhood relationships of certain vertices in the feature space, thereby preserving the topological structure of the feature space.

2) Replay-Based Methods: Replay-based methods allow the model to store a subset of data from previous tasks (referred to as exemplars) and revisit exemplars during the training phase of new tasks. Rainbow memory (RM) [43] suggested a method to assess the classification uncertainty of each sample and store those with a range of uncertainties to ensure a diverse sample set in memory. GDumb [44] proposed a method that involves the random removal of greedily stored samples from memory and highlights the importance of retaining old data. Region-aware memory (RAM) [45] proposed a method to divide the data distribution areas of each class into distinct regions using adversarial attacks, and selecting samples uniformly from these regions to ensure diversity. However, limited memory capacity restricts the amount of exemplars, and biased sampling affects the quality of exemplars, potentially leading to suboptimal performance.

In addition, feature replay offers an alternative that addresses limitations in storage. Some works have proposed generative data and generative feature methods, where generative adversarial networks (GAN) [46] is used for creating synthetic data or feature. Xiang et al. [47] suggested training a conditional adversarial network to synthesize features of old classes by using normalized feature embedding statistics that have been preserved. Tang et al. [48] introduced a learnable feature generator that utilizes unlabeled data to enrich the sample set in memory.

3) Dynamic Architecture-Based Methods: These methods provide a flexible framework for accommodating new tasks by adaptively expanding and pruning the model's architecture, which allows different parts of the model to specialize for distinct tasks. Piggyback [49] retained the parameters of a pretrained model and learns a mask to select the most suitable parameters for each task. PackNet [50] freed up model capacity by pruning and retraining to release unnecessary parameters after each task's training. Neural architecture search (NAS) [51] introduced a novel framework that separates neural structure optimization from parameter learning. The optimization component dynamically adapts the architecture for task accommodation, while the learning component fine-tunes the parameters. Progressive neural networks (PNN) [52] explicitly structured the transition between tasks by constructing a single-column network and leveraging the knowledge of previous tasks through horizontal connections. Dynamically expandable networks (DEN) [53] selectively updated the old network and dynamically enlarges its model size. Memory-efficient expandable model (MEMO) [54] extended specialized layers over shared generalized representations, effectively generating diverse representations with minimal overhead. Dynamically expandable representation (DER) [55] proposed a pruning strategy at the channel level using masks to expand the representation by leveraging exemplars and new data.

B. Class Imbalance and Bias Correction

Although techniques like exemplar memory and knowledge distillation have been employed to mitigate catastrophic forgetting, models face the limitation of prediction bias due to the imbalanced datasets, where old classes retain fewer samples than new classes. Javed and Shafait [56] emphasize that exemplar memory methods may result in imbalanced datasets, which can introduce bias in the final fully connected (FC) layer.

For replay-based methods, the imbalanced datasets result in smaller weight vectors and deviation from the original feature-to-weight vector relationships of old classes compared to new ones. Besides, it increases ambiguities that confound classification, leading to a classification score bias that skews predictions toward new classes. To address this issue, researchers have explored various strategies, including balanced training to ensure equal representation, equilibrium fine-tuning to recalibrate the model in each incremental phase, and postprocessing that adjusts model biases. End-to-end incremental learning (EEIL) [57] proposed a technique for fine-tuning the model using a balanced dataset after the initial training phase. UtilizIng gradient scaling by weighting losses (UW) [58] adopted a balanced fine-tuning strategy that scales gradients and weight losses according to training data statistics. Learning a unified classifier incrementally via rebalancing (LUCIR) [59] employed cosine normalization in the final layer

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to mitigate bias. Prototype augmentation and self-supervision (PASS) [60] introduced a method for augmenting the prototypes of old classes to alleviate bias in the FC layer. Weight aligning (WA) [61] integrated classifiers and aligned the logits output by the model for all class nodes. Slim et al. [62] introduced a method for obtaining bias correction parameters through dataset knowledge transfer on reference datasets.

C. Knowledge Distillation

Knowledge distillation [63] is a widely used technique in CIL that can alleviate catastrophic forgetting. Knowledge distillation involves transferring the knowledge from a teacher model to a student model, often by matching the soft labels of the teacher model with the output probabilities of the student model. This process helps the student model to retain the knowledge learned from previous tasks while adapting to new ones. Furthermore, distillation occurs not only in the output probability space but also in the feature representation space, enhancing the robustness of the model against forgetting. Additionally, relationships between layers, as quantified by techniques like the gram matrix, information flow, and singular value decomposition, are also distilled to minimize divergence between probability distributions of representations.

D. CIL Methods for RSSC

Recently, CIL methods in RSSC have garnered increasing attention and some works were proposed in applications. Ammour et al. [27] enhanced the distinction between the land cover classification task by learning from task prototypes kept in a linear memory. Convolutional neural network (CILEA-NET) [36] introduces a curriculum learning approach where the order of new classes is determined by their similarity to already trained classes. Training is then conducted in this specified order. Ye et al. [28] proposed the asymmetric collaborative network (SCN), which consists of a preserving network for retaining learned knowledge and a transient network for acquiring new information. Knowledge transfer is facilitated through triple distillation and prior feature fusion, enabling robust and adaptive performance across tasks. Lu et al. [29] proposed a lightweight incremental learning (LIL) network based on a dynamic architecture. It adapts the feature extractor for task-specific features, introduces a light feature transfer module for data distribution alignment, and employs a dual-constraint loss combining knowledge distillation and adversarial learning. Ammour [30] integrated data autogeneration subnetworks that sequentially train the model on distinct classes. One subnetwork minimizes discrimination errors between land cover classes during new task learning, while the other reproduces data from previous tasks by uncovering their latent structure. Ye et al. [31] introduced a channel attention-based multiscale depthwise network for RSSC. This network integrates effective channel attention and multiscale depthwise convolution for detailed feature extraction. The network adds new modules to adapt to new tasks and knowledge distillation further optimizes the model by reducing redundancy. Fu et al. [33] proposed an exemplar selection strategy integrating center and boundary information from prior classes and introduced a heterogeneous prototype-based framework to enhance exemplar informativeness.

In summary, most of the above methods effectively reduce catastrophic forgetting but still encounter several issues. First, replay strategy tends to favor new classes due to imbalanced data volumes, leading to higher prediction scores. There is a lack of systematic approach to mitigate this bias. Next, the use of SKD lacks proper justification and can introduce biases toward the recent tasks. In addition, most methods only concentrate on output probabilities rather than utilizing feature distillation at the same time. Furthermore, most methods overlook the rich multilevel semantics and multiscale features in RSIs, making it challenging to learn robust and rich feature representations.

III. Methodology

A. Problem Setting

In contrast to the traditional learning paradigm, where the model is trained on closed dataset, CIL requires the model to be trained on a sequence of tasks, each containing a new set of classes. The model must be able to classify all seen classes at the end of each stage.

In the tth incremental learning phase, also referred to as the task t, the training data includes data $\mathcal{D}_t$ from previously unseen new classes and exemplar memory $\mathcal{M}$. We define $\mathcal{D}_t = \{(\boldsymbol{x}_t^{(i)}, y_t^{(i)})\}_{i=1}^{n_t}$, where $\boldsymbol{x}_t^{(i)}$ is the input data for task t, $y_t^{(i)}$ denotes the corresponding target labels, and n_t is the number of training samples for new classes in task t. For each incremental task, we assume that a separate exemplar memory $\mathcal{M}$ is allocated to store data for old classes. Before the next phase, due to strict memory constraints, most samples in $\mathcal{D}_t$ are discarded, retaining only a few samples m in exemplar memory $\mathcal{M}$. These samples are selected from $\mathcal{D}_t$. Specifically, in the tth incremental learning phase ($t \geq 2$), we store m exemplars from each class that were learned until task $t-1$, where $\mathcal{M} = (t-1) \times m$. We assume $|\mathcal{M}| \ll n_t$.

In the tth incremental learning phase, our classification model can be denoted as θ_t, and the parameters of the model are learning by data $\mathcal{D}_t \cup \mathcal{M}$. The model is subsequently evaluated on a test set covering all classes $\sum_{j=1}^{t} C_j$. After training the model in the tth incremental learning phase, the class prediction for any test sample is obtained using the following formula:

$$\hat{y} = \underset{y \in \sum_{j=1}^{t} C_j}{\arg\max}\ z_{ty}(x, \theta_t) \tag{1}$$

where z_{ty} is the output score of the model θ_t for class $y \in \sum_{j=1}^{t} C_j$ before the softmax function. In testing, the final FC layers are combined, and predictions are formed for all possible classes $\sum_{j=1}^{t} C_j$.

B. Overview

To tackle catastrophic forgetting and strike a balance between stability and plasticity, we introduce CBCNet, an effective CIL framework for RSSC. The overall structure of the network, along with the detailed schematics of CIAM and CIFM, is illustrated in Fig. 2. The structure is composed of an input layer, three stages of residual blocks augmented with CIAM and CIFM, a global average pooling layer, and a final

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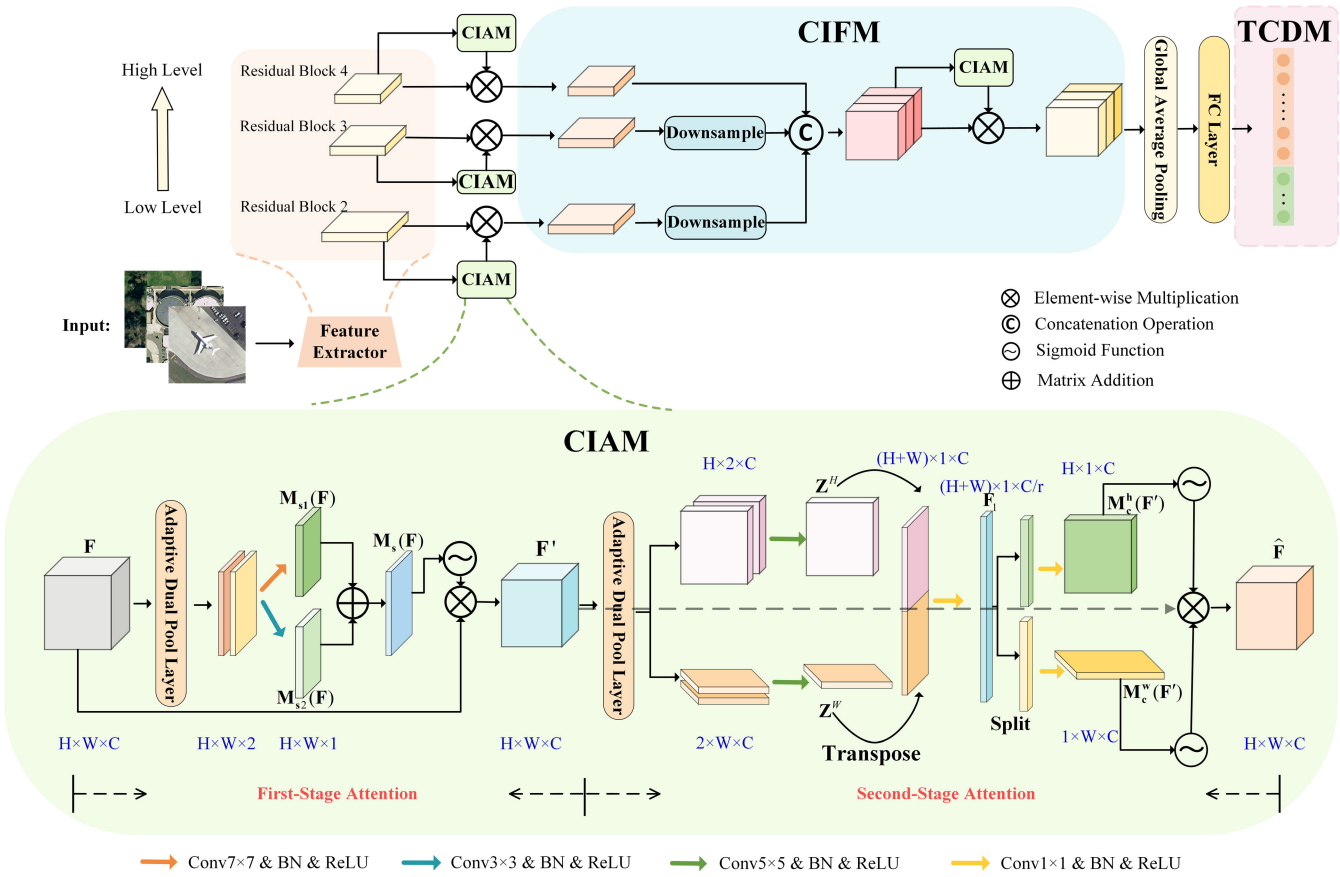


Fig. 2. Overall structure of the network (CBCNet), along with the detailed schematic of CIAM and CIFM. In CBCNet, the last three stages of residual blocks are augmented with CIAM and CIFM, and the final FC layer has been modified by TCDM. In CIAM, the input and output sizes of the key steps are listed.

FC layer that has been modified by TCDM. The original RSIs are fed into the feature extractor as input, and then processed through the residual blocks. Residual blocks capture multi-level features: low-level features capture image fundamentals, middle-level features incorporate contextual information, and high-level features capture rich semantic information. To leverage these distinct characteristics, we select the outputs of the last layer from the final three stages of residual blocks. Then, CIAM and CIFM are employed for feature extraction. In CIAM, we factorize 2-D global pooling into three 1-D feature encoding operations using the adaptive dual pool layer (D-pool layer) to encode each dimension along the H-, W-, and C-axis. This enables us to capture cross-dimensional dependencies while preserving position-sensitive information. In CIFM, we adopt a strategy to ensure effective interaction between convolutional features at different levels. The three feature maps are concatenated to yield the aggregated feature map, which is subsequently processed through the CIAM to obtain the final aggregated feature, incorporating multilevel information. Then the output feature map is fed into a global average pooling layer, followed by an FC layer that has been modified by TCDM.

C. Cross-Dimensional and Interaction-Aware Attention Mechanism

Benefiting from the ability to construct interchannel dependencies, squeeze-and-excitation (SE) attention and its variants have been widely applied in RSSC. Despite their merits, the following issues exist. First, these methods often overlook spatial information, which is crucial for a holistic understanding of the RS scenes. Additionally, channel attention mechanisms typically convert a feature tensor into a single feature vector through 2-D global pooling, which leads to a lack of position sensitivity and local detail information. Furthermore, these methods compute attention maps along a single dimension, neglecting the importance of capturing cross-dimensional interactions, which we demonstrate to have a positive impact on performance.

To help models more accurately locate and recognize objects of interest in RSIs, we present CIAM to capture CIAM maps that effectively highlights informative regions and suppresses unnecessary ones within images, thereby creating an effective feature descriptor. As shown in Fig. 2, we factorize the 2-D global pooling used in SE attention into three 1-D feature encoding operations, utilizing the adaptive D-pool layer to encode each dimension along the H-, W-, and C-axis, respectively. This allows us to capture dependencies along one specific dimension while preserving position-sensitive information along the other two dimensions. The resulting dependencies are then encoded separately into three attention maps that incorporate channel, position, and direction-aware information. Moreover, CIAM is lightweight and flexible, making it easily integrable into existing network architectures.

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1) Adaptive D-Pool Layer: The D-pool layer, which refers to the adaptive dual-pool layer, is a novel component designed to reduce a certain dimension of the tensor to two. This is achieved through the concatenation of both average-pooled and max-pooled features across the targeted dimension. Traditional mixed pooling techniques combine the outputs of max and average pooling through averaging or weighted sums, whereas our approach concatenates these features into two channels, preserving their distinct characteristics. In this manner, the layer can significantly reduce the depth while maintaining a rich representation of the original features. This reduction facilitates more efficient computations in subsequent layers. Mathematically, it can be represented by the following equation:

$$\text{D-pool}_{i_d}(\mathbf{F}) = \text{Concat}\left[\text{MaxPool}_{i_d}(\mathbf{F}), \text{AvgPool}_{i_d}(\mathbf{F})\right] \quad (2)$$

where $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ is the intermediate feature map, and i_d is the ith dimension ($i = 1, 2, 3$) where the max and average pooling operations take place. For instance, the D-pool of a tensor of shape ($C \times H \times W$) in the first dimension yields a ($2 \times H \times W$) tensor.

CIAM takes a specific feature tensor $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ as input and outputs a transformed tensor of the same size, with augmented representations. CIAM consists of two sequential submodules, forming the attention map through two stages. We describe the detailed operation below.

2) First-Stage Attention: First, D-pool operations are applied along the channel axis, producing a concatenated set of pooled features. To get more effective multiscale feature map, it is passed through a parallel $f^{3\times3}$ and $f^{7\times7}$ convolution layer, compressing the channel depth to 1. Subsequently, the two output feature vectors are combined through matrix addition to generate a spatial attention map. The final attention map is normalized using a sigmoid function σ to scale attention values between 0 and 1, thereby distinguishing foreground from background regions clearly.

In short, the first-stage attention map $\mathbf{M_s}(\mathbf{F})$ can be formulated as follows:

$$\mathbf{M_{s1}}(\mathbf{F}) = \sigma\left(f^{7\times7}\left(\text{D-pool}_{1_d}(\mathbf{F})\right)\right) \quad (3)$$

$$\mathbf{M_{s2}}(\mathbf{F}) = \sigma\left(f^{3\times3}\left(\text{D-pool}_{1_d}(\mathbf{F})\right)\right) \quad (4)$$

$$\mathbf{M_s}(\mathbf{F}) = \mathbf{M_{s1}}(\mathbf{F}) \oplus \mathbf{M_{s2}}(\mathbf{F}) \quad (5)$$

where σ denotes the sigmoid function, and $f^{7\times7}$ and $f^{3\times3}$ represent convolution operations with filter sizes of 7×7 and 3×3, respectively. The $\oplus$ symbol indicates matrix addition.

The first-stage attention $\mathbf{M_s}(\mathbf{F})$ captures cross-dimensional and strong interaction dependencies between the H- and W-axis, representing complex spatial relationships and extracting multiscale spatial features. The output can be calculated as follows:

$$\mathbf{F}' = \mathbf{M_s}(\mathbf{F}) \otimes \mathbf{F} \quad (6)$$

where $\otimes$ denotes elementwise multiplication.

3) Second-Stage Attention: To capture the strong interaction dependencies between other dimensions, D-pool operations are applied along the H- and W-axis, respectively. And then they are passed through a convolution layer $f^{5\times5}$ separately to compress the horizontal depth and vertical depth to 1. This yields a pair of direction-aware feature maps $\mathbf{Z}^H$ and $\mathbf{Z}^W$ that capture cross-dimensional and strong-interaction dependencies along one spatial direction while preserving precise positional information along the other, which can be formulated as follows:

$$\mathbf{Z}^H = f^{5\times5}\left(\text{D-pool}_{2_d}(\mathbf{F}')\right) \quad (7)$$

$$\mathbf{Z}^W = f^{5\times5}\left(\text{D-pool}_{3_d}(\mathbf{F}')\right) \quad (8)$$

where $\mathbf{F}' \in \mathbb{R}^{C\times H\times W}$ is the intermediate feature map, and 2_d and 3_d are the second dimension and the third dimension where the D-pool operations take place.

By applying a transpose operation to $\mathbf{Z}^W$ along the C-axis, we transform its shape from ($1 \times W \times C$) to ($W \times 1 \times C$). Then, $\mathbf{Z}^H$ and the transposed $\mathbf{Z}^W$ are concatenated along the first dimension and passed through a shared $f^{1\times1}$ convolution, producing an intermediate feature map $\mathbf{F}_1$, which can be formulated as follows:

$$\mathbf{F}_1 = \delta\left(f^{1\times1}\left(\text{Concat}\left[\mathbf{Z}^H, \left(\mathbf{Z}^W\right)^{\text{T}}\right]\right)\right) \quad (9)$$

where δ is a nonlinear activation function and the concatenation occurs along the spatial dimension.

$\mathbf{F}_1 \in \mathbb{R}^{C/r\times(H+W)}$ is the intermediate feature map that encodes spatial information in both horizontal and vertical directions. This process facilitates the alignment of features from the spatial positions of both dimensions. The transpose operation brings features from different dimensions into a common space, enabling the computation of a unified attention map that incorporates both horizontal and vertical spatial dependencies, thereby fostering richer interactions between them. To manage model complexity, the channel number is reduced with an appropriate reduction ratio r.

Subsequently, $\mathbf{F}_1$ is split along the spatial dimension into two tensors $\mathbf{F}_1^h \in \mathbb{R}^{C/r\times H}$ and $\mathbf{F}_1^w \in \mathbb{R}^{C/r\times W}$. Two separate convolutional transformations $f_h^{1\times1}$ and $f_w^{1\times1}$ are applied to $\mathbf{F}_1^h$ and $\mathbf{F}_1^w$ to yield tensors with the same channel number as the input feature, which can be expressed as follows:

$$\mathbf{M_c^h}(\mathbf{F}') = \sigma\left(f_h^{1\times1}(\mathbf{F}_1^h)\right) \quad (10)$$

$$\mathbf{M_c^w}(\mathbf{F}') = \sigma\left(f_w^{1\times1}(\mathbf{F}_1^w)\right). \quad (11)$$

This process integrates positional information into channel attention via two cross-channel encoding steps, aggregating features both vertically and horizontally. This unique process generates a pair of cross-channel, direction-aware, and position-sensitive attention maps, which are applied to augment feature representations. The outputs $\mathbf{M_c^h}(\mathbf{F}') \in \mathbb{R}^{C\times H\times1}$ and $\mathbf{M_c^w}(\mathbf{F}') \in \mathbb{R}^{C\times1\times W}$ are then used as the second-stage attention weights.

CIAM includes two sequential submodules that generate the attention map in two stages. The overall process of CIAM can be summarized as follows:

$$\mathbf{F}' = \mathbf{M_s}(\mathbf{F}) \otimes \mathbf{F} \quad (12)$$

$$\hat{\mathbf{F}} = \mathbf{M_c^h}(\mathbf{F}') \otimes \mathbf{M_c^w}(\mathbf{F}') \otimes \mathbf{F}' \quad (13)$$

where $\hat{\mathbf{F}}$ is the final refined feature map. CIAM generates cross-dimensional attention maps that integrate channel, position, and direction-aware information, thereby enriching the input feature representation.

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D. Contextual Information Fusion Module

The self-attention mechanism in CIAM is effective at capturing context within a single level of features. CIAM focuses on intralevel interactions (e.g., spatial and channelwise dependencies within a single feature map) but does not explicitly capture cross-level interactions. To aggregate information from multiple levels and better capture contextual information across multiple scales, we propose a novel CIFM. CIFM captures rich contextual information by exploring correlations among multilevel features, thereby consolidating low-level pixel information from RSIs and constructing a more advanced semantic understanding.

ResNet18 is a lightweight ResNet variant designed to address vanishing gradients in deep networks using residual blocks. These blocks capture multilevel features at different stages. As shown in Fig. 2, to combine the distinct characteristics of features at different levels, we select the outputs of the last layer from the final three stages of residual blocks, respectively representing low-level, mid-level, and high-level features. For ease of description, these are denoted as $\mathbf{F}_1 \in \mathbb{R}^{H_1\times W_1\times C_1}$, $\mathbf{F}_2 \in \mathbb{R}^{H_2\times W_2\times C_2}$, and $\mathbf{F}_3 \in \mathbb{R}^{H_3\times W_3\times C_3}$.

CIAM highlights informative regions within images, thereby creating an effective feature descriptor. In CIFM, CIAM is cascaded to augment feature representations and feature fusion. Specifically, $\mathbf{F}_1, \mathbf{F}_2$, and $\mathbf{F}_3$ are fed into the CIAM to obtain the enhanced feature representations $\hat{\mathbf{F}}_1, \hat{\mathbf{F}}_2$, and $\hat{\mathbf{F}}_3$, which contain local textures information, complex contextual information, and rich semantic structures within RS scenes.

We adopted a strategy to ensure effective interaction between convolutional features at different levels. First, average pooling reduced the spatial resolution of $\hat{\mathbf{F}}_1$ and $\hat{\mathbf{F}}_2$. Subsequently, 1×1 convolution operations were used to unify their channel numbers. This transformation resulted in new feature representations $\hat{\mathbf{F}}'_1$ and $\hat{\mathbf{F}}'_2$ for $\hat{\mathbf{F}}_1$ and $\hat{\mathbf{F}}_2$, respectively, both with the shape of $\mathbb{R}^{H_3\times W_3\times C_3}$, matching that of the higher level feature $\hat{\mathbf{F}}_3$. Then, they are concatenated along the channel dimension to yield the aggregated feature map as follows:

$$\mathbf{Y} = \text{Concat}\left[\hat{\mathbf{F}}'_1, \hat{\mathbf{F}}'_2, \hat{\mathbf{F}}_3\right]. \tag{14}$$

Subsequently, the feature is sent through the CIAM to obtain the final aggregated feature $\hat{\mathbf{Y}}$, which incorporates multilevel information. CIFM integrates adaptive feature modulation through attention mechanism, enhancing its ability to dynamically adjust the importance of different feature levels and balancing the representation of detailed textures and semantic concepts. The attention-based refinement and unique feature fusion strategy in CIFM distinguish it from other conventional multilevel feature fusion techniques, such as feature pyramid networks (FPN) [64]. The process is detailed as follows:

$$\mathbf{Y}' = \mathbf{M_s}(\mathbf{Y}) \otimes \mathbf{Y} \tag{15}$$

$$\hat{\mathbf{Y}} = \mathbf{M_c^h}\left(\mathbf{Y}'\right) \otimes \mathbf{M_c^w}\left(\mathbf{Y}'\right) \otimes \mathbf{Y}' \tag{16}$$

where $\hat{\mathbf{Y}}$ is the final refined feature map. CIFM captures rich contextual information by exploring correlations among multilevel features, and balances the low-level details with the high-level semantics through attention-based refinement, thereby consolidating low-level pixel information from RSIs and constructing a more advanced semantic understanding.

E. Taskwise Classifier Head Decoupling Mechanism

Replay-based methods introduce a significant bias toward the new classes owing to data imbalance, which is manifested in the higher logit values for new classes when compared to old classes. Within the framework of CIL, a significant challenge emerges from the application of standard CE loss. The specific reasons will be explained below.

In the tth incremental learning phase, the CE loss for a sample $(\mathbf{x}, y)$ can be expressed as follows:

$$L_{\text{CE},t}((\mathbf{x}, y), \theta) = -\sum_{c=1}^{C_t} y_c \log\left(p_{1:t,c}(\mathbf{x}, \theta)\right) \tag{17}$$

where θ denotes the model parameters. y_c is a one-hot encoded vector where $y_c = 1$ only when $c = y$, and $y_c = 0$ otherwise. In this case, $p_{1:t,c}(\mathbf{x}, \theta)$ denotes the cth component of the probability vector computed by using all output scores from task 1 to task t, which is calculated as follows:

$$p_{1:t,c}(\mathbf{x}, \theta) = \frac{e^{z_{\text{tc}}(\mathbf{x},\theta)}}{\sum_{k\in C_{1:t}} e^{z_{tk}(\mathbf{x},\theta)}}. \tag{18}$$

To systematically analyze the root causes of prediction bias toward new classes, we need to calculate the gradient of the CE loss $L_{\text{CE},t}$ with respect to the logits z_{tc}. This gradient reflects how the model adjusts its internal weights to minimize the loss function. First, according to the chain rule, it is calculated as follows:

$$\frac{\partial \mathcal{L}_{\text{CE},t}((\mathbf{x}, y), \theta)}{\partial z_{tc}} = -\sum_{c=1}^{C_t} y_c \cdot \frac{\partial}{\partial z_{\text{tc}}} \log\left(p_{1:t,c}(\mathbf{x}, \theta)\right). \tag{19}$$

Since $y_c = 0$ when $c \neq y$, only the term corresponding to $c = y$ contributes to the sum. $p_{1:t,c}(\mathbf{x}, \theta)$ is the result of the softmax function and the derivative of it can be calculated as follows:

$$\frac{\partial}{\partial z_{ty}} \log(p_{1:t,y}(\mathbf{x}, \theta)) = \frac{1}{p_{1:t,y}(\mathbf{x}, \theta)} \cdot \frac{\partial p_{1:t,y}(\mathbf{x}, \theta)}{\partial z_{ty}}. \tag{20}$$

For $c \neq y$, $y_c = 0$ and $y_y = 1$. Similarly, for $c = y$, $y_c = 1$. Therefore, the gradient of the CE loss with respect to the output scores can be calculated as follows:

$$\frac{\partial \mathcal{L}_{\text{CE},t}((\mathbf{x}, y), \theta)}{\partial z_{\text{tc}}} = p_{1:t,c}(\mathbf{x}, \theta) - \mathbb{1}\{y = c\}. \tag{21}$$

Note that the indicator function $\mathbb{1}\{y = c\}$ returns 1 if class c equals the true label y, and 0 otherwise. Specifically, (21) is always positive for $c \neq y$, which means the gradient of the classification scores tends to be predominantly positive for old classes, resulting in an overpenalization of output scores for old classes. Therefore, the classification scores for old classes tend to decline. The uneven distribution of gradient updates in the presence of imbalanced datasets is a critical issue in CIL. As illustrated in Fig. 3, the process begins with an imbalanced dataset $\mathcal{D}_t \cup \mathcal{M}$. During the training process, the classification scores for the old classes tend to decrease, while those for new classes increase. There is an uneven distribution of gradient updates, skewing the classification scores toward the new classes. Thus, we believe that the imbalanced gradient descent steps for the classification scores of old classes result in a significant bias toward the new classes. This issue stems

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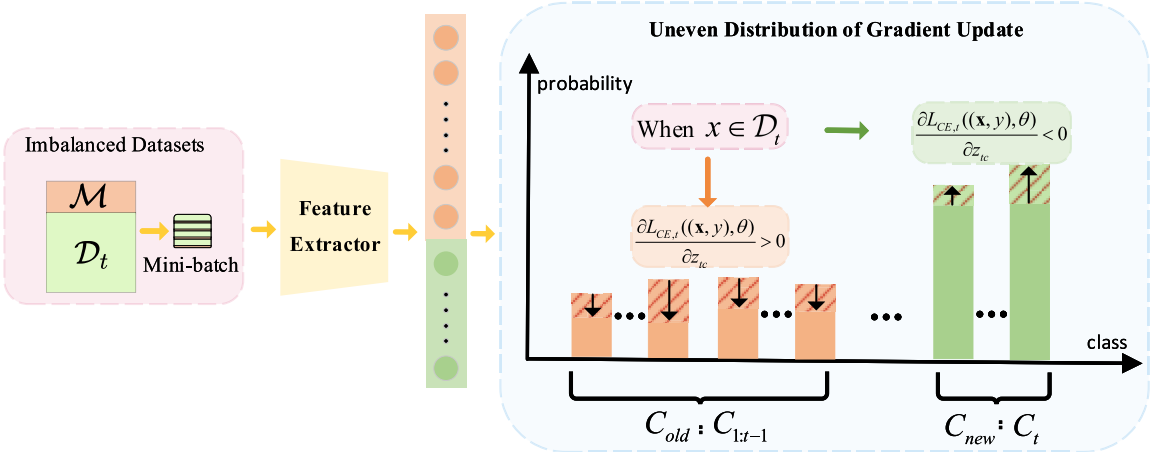


Fig. 3. Illustration of the uneven distribution of gradient update. For the abundant samples $x \in \mathcal{D}_t$, the classification scores for the old classes will continue to decrease during the gradient descent steps, while those for new classes will increase. When models are updated using an imbalance dataset $\mathcal{D}_t \cup \mathcal{M}$, there is an uneven distribution of gradient updates, skewing the classification scores toward the new classes.

from the combination of classifier head for new and old classes when computing prediction probabilities.

To solve the above problems, we propose a novel TCDM. As new tasks arrive, the classifier head dynamically expands and then decouples new class heads from old ones. The core idea is to achieve a gradient-balanced mechanism by decoupling the classifier head, ensuring that classification scores of old classes do not continuously decline during the gradient descent steps for new classes. Furthermore, an adaptive gradient-balanced CE loss is introduced to compensate for the imbalanced gradient contributions to the classification scores. For clarity, in the tth incremental learning phase, let C_{old} be the classes from the previous tasks, and C_{new} be the new classes introduced in the current task. The output logits from the current model are split into two segments: those corresponding to the old classes and those for the new classes.

The adaptive gradient-balanced CE loss $\mathcal{L}_{\text{GB-CE}}$ is defined as follows:

$$\mathcal{L}_{\text{GB-CE}}((x,y),\theta) = -\sum_{c\in C_{\text{old}}} \mathbb{1}\{y=c\}\log p_{1:t-1,c}(x,\theta) + D_{\text{KL}}(y_t||p_t(x,\theta))\cdot\mathbb{1}\{y\in C_{\text{new}}\} \quad (22)$$

where $p_{1:t-1,c}(x,\theta)$ denotes the prediction probability for C_{old}, and y_t represents the one-hot vector in $\mathbb{R}^{C_{\text{new}}}$. $p_t(x,\theta)$ denotes the prediction probability for C_{new} and $D_{\text{KL}}(\cdot|\cdot)$ represents the Kullback–Leibler divergence.

The prediction probabilities are calculated independently using just the output scores for C_{old} or C_{new}, and the CE loss is evaluated separately as well. It now becomes evident that when $(x,y)\in\mathcal{D}_t$ and $c\in C_{\text{old}}$, $(\partial L_{\text{CE-SS}})/(\partial z_{tc})=0$. As a result, the gradient contributions from samples in C_{new} will not excessively impact the classification scores for C_{old}. Therefore, TCDM constrains prediction bias toward new classes and enhances the model's discrimination among all seen classes.

F. Multilevel Integrated Knowledge Distillation Module

Typically, most replay-based methods in CIL emphasize constraining the model's performance change for earlier tasks within the output probability space, while ignoring the feature representation space. To address this limitation, we propose a novel MKDM to ensure comprehensive knowledge transfer. Specifically, MKDM encompasses TWD in output probability space and HPD in feature representation space.

1) Taskwise Knowledge Distillation: SKD in output probability space has been widely adopted to retain knowledge from past tasks. For an input data x, the distillation loss defined for SKD when learning task t is calculated as follows:

$$\mathcal{L}_{\text{SKD},t}(\mathbf{x},\theta) = D_{\text{KL}}\left(p^{\tau}_{1:t-1}(\mathbf{x},\theta_{t-1})||p^{\tau}_{1:t-1}(\mathbf{x},\theta_t)\right) \quad (23)$$

where $D_{\text{KL}}(\cdot|\cdot)$ represents the Kullback–Leibler divergence, τ signifies the temperature scaling parameter, and θ_{t-1} and θ_t represent the model parameters in the $(t-1)$th and tth incremental learning phases, respectively.

However, this approach can inadvertently preserve biases introduced by the CE loss function, with the similar reason as mentioned in Section III-E. Specifically, it will introduce a significant bias in the soft targets, skewing them toward the most recent old tasks. When these biased soft targets are used in the distillation loss, they can exacerbate the bias.

To address this limitation, we adopt TWD in output probability space, which employs separate soft targets computations for each task, making it less susceptible to taskwise biases. For an input data x, the distillation loss defined for TKD when learning task t is as follows:

$$\mathcal{L}_{\text{TWD},t}(\mathbf{x},\theta) = \sum_{s=1}^{t-1} D_{\text{KL}}\left(p^{\tau}_{s}\left(\mathbf{x},\theta_{t-1}||p^{\tau}_{s}(\mathbf{x},\theta_t)\right)\right). \quad (24)$$

Furthermore, in (23) and (24), we detail the cth component of the probability vectors $p^{\tau}_{s}(\mathbf{x},\theta)$ and $p^{\tau}_{1:s}(\mathbf{x},\theta)$ calculated in STD and TKD

$$p^{\tau}_{s,c}(\mathbf{x},\theta) = \frac{e^{z_{sc}(\mathbf{x},\theta)/\tau}}{\sum_{k\in C_s} e^{z_{sk}(\mathbf{x},\theta)/\tau}} \quad (25)$$

$$p^{\tau}_{1:s,c}(\mathbf{x},\theta) = \frac{e^{z_{sc}(\mathbf{x},\theta)/\tau}}{\sum_{k\in C_{1:s}} e^{z_{sk}(\mathbf{x},\theta)/\tau}}. \quad (26)$$

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Minimizing either (23) or (24) leads to regularizing the old model θ_{t-1}. By comparing the two probability vectors formally, $p^{\tau}_{1:t-1}(\mathbf{x}, \theta_{t-1})$ uses the global prediction probabilities across all previous tasks, while $\{p^{\tau}_{s}(\mathbf{x}, \theta_{t-1})\}_{s=1}^{t-1}$ employs the task-specific prediction probabilities, obtained separately for each task. Moreover, TWD is particularly suitable for scenarios where TCDM is used, as it does not suffer from the same biases when computing the classification scores.

2) High-Plasticity Knowledge Distillation: Most replay-based methods in CIL overlook the critical role of knowledge distillation within feature representation space. Therefore, our solution involves augmenting the replay memory with activations from intermediate layers to encourage the model to capture discriminative information, similar to the initial model. While storing intermediate activations can indeed be resource-intensive, we have implemented several optimization strategies to mitigate this issue. Specifically, we employ floating-point precision reduction to efficiently store the activations. Furthermore, we constrain the output feature map of CIFM to implement the knowledge distillation in the feature representation space. By aggregating the channels of intermediate layers, we retain only the spatial coordinates, significantly reducing memory overhead while preserving essential information. The knowledge distillation loss in HPD is defined as follows:

$$\mathcal{L}_{\text{HPD}} = \sum_{w=1}^{W}\sum_{h=1}^{H}\left\|\sum_{c=1}^{C}\mathbf{h}^{t-1}_{\ell,c,w,h} - \sum_{c=1}^{C}\mathbf{h}^{t}_{\ell,c,w,h}\right\|^2. \tag{27}$$

We use l^2-normalized Euclidean distance, $||\cdot - \cdot||$, to constrain feature drift for a certain intermediate feature layer.

By aggregating the channels, only the spatial coordinates are retained. This allows the knowledge distillation to be more robust to changes in the input data, as it can adapt by shifting the representation of certain features to different channels, thus allowing reorganization across channels. Without pooling, the network must learn to accurately represent the input data at every spatial location, leading to the most stringent loss function, the most strict constraints, and the lowest level of plasticity.

G. Algorithm Scheme

As shown in Fig. 4, by combining classification loss and dual-constraint distillation loss, the overall loss for CBCNet is given as follows:

$$\mathcal{L} = \mathcal{L}_{\text{GB-CE}} + \lambda_1\mathcal{L}_{\text{TWD}} + \lambda_2\mathcal{L}_{\text{HPD}} \tag{28}$$

where λ_1 and λ_2 are balance factors to adjust the contribution of each loss component.

The pseudocode for the training procedure during the tth incremental learning phase ($t \geq 2$) is shown in the algorithm.

Algorithm 1 Pseudocode for Our Method Training (tth Phase, $t \geq 2$)

Input: New data $\mathcal{D}_t = \left\{(\boldsymbol{x}_t^{(i)}, y_t^{(i)})\right\}_{i=1}^{n_t}$ and exemplar memory $\mathcal{M}$ of old classes for the task t; the old model F_{t-1} with weights θ_{t-1} performed well in the $t-1$ th incremental learning phase; hyperparameters λ_1 and λ_2.
Output: The updated weights θ_t of the model F_t that can classify both new and old tasks.

Initialize new module F_t with θ_{t-1} and extend the classifier head for new classes.
for i = 1 to the number of epochs **do**
  Calculate $\mathcal{L}_{\text{GB-CE}}$ using the outputs of the new model F_t, according to (22).
  Calculate $\mathcal{L}_{\text{TWD}}$ using the outputs of both the new model F_t and the old model F_{t-1}, according to (24).
  Calculate $\mathcal{L}_{\text{HPD}}$ using the outputs of both the new model F_t and the old model F_{t-1}, according to (27).
  $\mathcal{L} = \mathcal{L}_{\text{GB-CE}} + \lambda_1\mathcal{L}_{\text{TWD}} + \lambda_2\mathcal{L}_{\text{HPD}}$.
  Update $F_t \leftarrow \theta_t - \frac{\partial \mathcal{L}}{\partial \theta_t}$.
end for
Retain the new model as F_t with θ_t
Select exemplars m from $\mathcal{D}_t$ to update the exemplar memory $\mathcal{M}$ by herding [26].
Discard the old model F_{t-1}.

IV. Experiments and Results

In this section, we conduct comprehensive experiments to assess the performance of CBCNet. First, we detail the experimental setting, including the five datasets used in our experiments, the selected baseline methods for comparison, the evaluation metrics, and the implementation details of all compared methods. Next, we perform sequential learning experiments to analyze how the model's performance changes with an increasing number of tasks. Then, we analyze how the number of selected exemplars affects model performance. Finally, we conduct ablation studies to clarify the role of each module within our method and assess its robustness to variations in hyperparameters.

A. Experimental Setting

1) Datasets: We selected five remote sensing datasets for our experiments: Aerial Image Dataset (AID) [65], RSI Classification Benchmark with 256 classes (RSI-CB256) [66], Northwestern Polytechnical University 45-class dataset (NWPU-45) [67], UC-Merced dataset [68], and PatternNet dataset [69]. Specifically, the AID, RSI-CB256, and NWPU-45 datasets were utilized for algorithm comparisons with the selected baseline methods. The NWPU-45 dataset was used for sequential learning analysis, while RSI-CB256 was employed to investigate the impact of the number of selected exemplars. The UC-Merced dataset was used to study the effect of hyperparameters. Notably, all five datasets were employed in ablation studies to verify the role of each novel component of CBCNet in complex CIL scenarios.

Specifically, we set five different CIL scenarios: six tasks with five classes for AID, seven tasks with five classes for RSICB-256, nine tasks with five classes for NWPU-45, six tasks with six classes for PatternNet, and three tasks with seven classes for UC-Merced.

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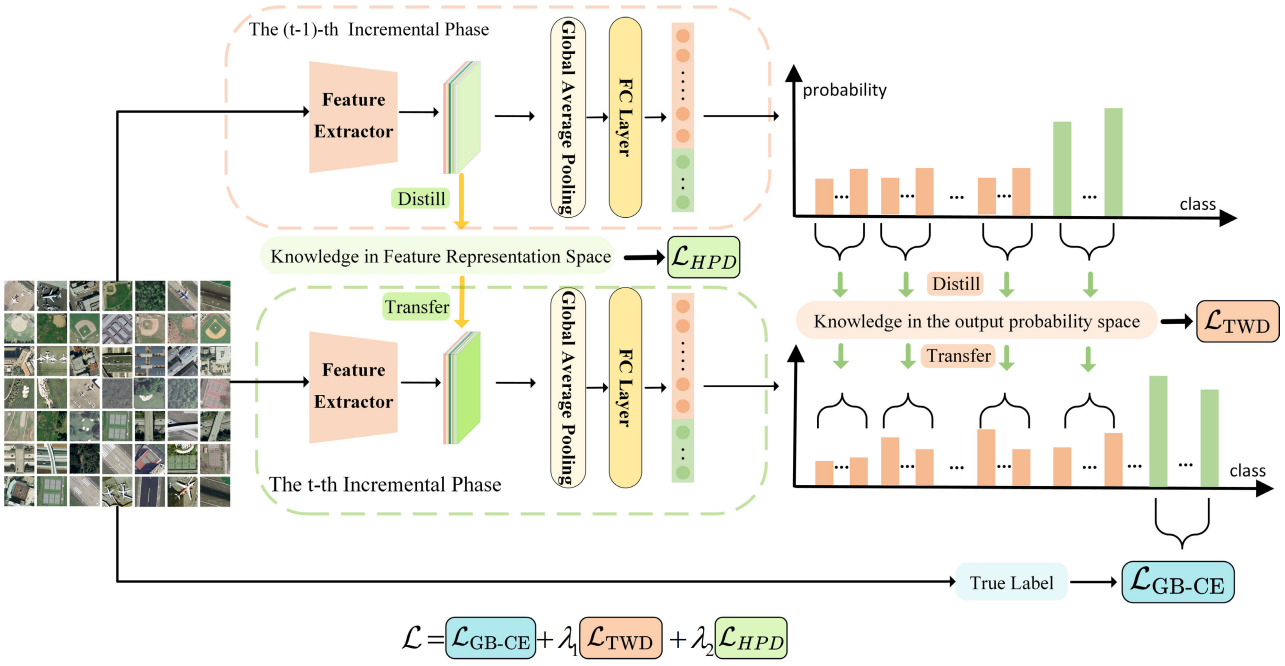


Fig. 4. Illustration of the overall loss for CBCNet. $\mathcal{L}_{\text{GB-CE}}$ compensates for imbalanced gradient contributions to classification scores. $\mathcal{L}_{\text{TWD}}$ employs separate soft targets computations for each task, reducing susceptibility to taskwise biases. $\mathcal{L}_{\text{HPD}}$ augments the replay memory with intermediate layer activations. λ_1 and λ_2 are balance factors to adjust the contribution of each loss component.

The characteristics of these datasets were listed as follows.

1) The AID dataset consists of 30 types of scene classes, with over 1000 images per class. Standardized to 600 × 600 pixels, the images cover resolutions from 8 to 0.5 m. The AID dataset's high intraclass variability, including significant scale, style, and imaging condition differences, poses challenges to RCCS.
2) The RSI-CB256 dataset consists of 35 types of scene classes, with approximately 690 images per class, totaling over 24 000 images. The images are standardized to 256 × 256 pixels, with each class featuring diverse angles, scales, shapes, and colors to enhance diversity.
3) The NWPU-45 dataset consists of 45 types of scene classes, with 700 images per class, totaling 31 500 images. Standardized to 256 × 256 pixels, the images vary widely in spatial resolution (30 to 0.2 m per pixel) and exhibit significant variability in translation, viewpoint, orientation, lighting, background, and occlusion.
4) The PatternNet dataset consists of 38 types of scene classes, with 800 images per class, totaling over 30 000 images. Standardized to 256 × 256 pixels, the images have spatial resolutions ranging from 0.062 to 4.693 m per pixel, showing rich diversity. To ensure consistent class numbers across incremental phases, the airplane and baseball field classes were removed.
5) The UC-Merced dataset consists of 21 types of scene classes, with 100 images per class, totaling over 2100 images. All images are standardized to 256 × 256 pixels with a uniform resolution of approximately 0.3 m, and the dataset shows significant variability in land use.

2) Baselines: Nine classical incremental learning methods were employed to compare with our method.

a) *Joint:* This is considered the ideal scenario, as it combines all available data for training simultaneously. It can be regarded as the performance ceiling for CIL methods.
b) *Fine-tuning:* This serves as the baseline for CIL, focusing on new classes but experiencing severe catastrophic forgetting.
c) *GEM [40]:* This computes the gradient of the model on memory set and the current task and projects the gradient onto the direction that minimizes the angle with the memory set.
d) *EWC [22]:* This is the first to propose using parameter regularization to prevent catastrophic forgetting. It utilizes the Fisher information matrix to constrain changes to important parameters.
e) *LWF [41]:* This is a representative method in distillation-based method, which incorporates a distillation loss into the overall loss function to prevent catastrophic forgetting.
f) *iCaRL [26]:* This employs both knowledge distillation loss and exemplar memory to avoid catastrophic forgetting and computes class prototypes for each observed class for classification.
g) *WA [61]:* This integrates classifiers learned at different incremental stages and aligns the logits output by the model for both old and new class nodes.
h) *DER [55]:* This employs a dynamically expandable architecture and a two-stage learning. At each step, the existing representation is frozen and augmented with new feature dimensions. Channel-level mask-based pruning dynamically adjusts the representation, and an auxiliary loss enhances discriminative capability for new classes.

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT METHODS FOR CIL ON THREE DATASETS

Method \ Dataset	AID (6 tasks with 5 classes)		RSI-CB256 (7 tasks with 5 classes)		NWPU-45 (9 tasks with 5 classes)	
	mACC(%)	BWT(%) ↓	mACC(%)	BWT(%) ↓	mACC(%)	BWT(%) ↓
Joint	95.31	-	96.42	-	87.69	-
Finetuning	20.46	81.27	19.45	85.24	10.47	92.73
GEM	27.47	76.92	21.09	82.09	8.16	94.14
EWC	26.52	74.19	23.37	81.24	12.72	89.27
LwF	51.26	50.18	46.24	57.62	26.82	76.37
iCaRL	63.85	39.76	67.93	29.68	46.28	49.02
WA	69.29	31.08	71.82	24.92	54.53	45.13
SCN	86.15	10.97	83.97	14.74	76.16	32.17
DER	78.32	21.06	87.74	13.15	77.28	29.97
CBCNet(ours)	**89.27**	**9.29**	**90.68**	**10.17**	**80.31**	**19.61**

i) *SCN [28]:* This addresses the challenge of catastrophic forgetting in RSSC by employing dual subnetworks. They collaborate through a triple distillation mechanism and a recall filter. Trans-Net learns new tasks effectively with attention mechanism, while Pres-Net preserves historical knowledge.

3) Evaluation Metrics: The model is evaluated on all previously learned tasks after training each new task. Referring to [19], three metrics are leveraged to evaluate the performance of algorithms: accuracy, mean accuracy (mACC), and backward transfer (BWT).

a) Accuracy: Accuracy measures the percentage of correct predictions out of the total number of predictions, serving as an indicator of an algorithm's performance on a single task

$$\text{Accuracy}(T) = \frac{\text{Num}\left(\hat{y} = y\right)}{\text{Num}(y)} \tag{29}$$

where T represents the tth task, $\hat{y}$ represents the model's output for the tth task, and y represents the true label.

b) mACC: The mACC is an important measure of the performance of multiclass classification models. Notably, the calculation of mACC involves averaging the model's accuracy on each task after the entire sequence of tasks has been learned. mACC is calculated as follows:

$$\text{mACC}(i) = \frac{1}{T}\sum_{i=1}^{T} P_{T,i} \tag{30}$$

where T represents the total count of tasks learned by the model, and $P_{T,i}$ represents the accuracy achieved in learning the ith task when all the tasks have been learned.

c) BWT: The BWT is used to measure the impact of learning new tasks on the performance of the previously learned tasks. BWT is calculated as follows:

$$\text{BWT}(T) = \frac{1}{T-1}\sum_{i=1}^{T-1}\left(P_{i,i} - P_{T,i}\right). \tag{31}$$

A positive BWT value suggests forgetting of the old tasks after learning the new ones, whereas negative BWT indicates that learning new task has improved the performance on the old tasks.

4) Implementation Details: ResNet-18 was chosen as the backbone network for all methods, and the data was split into training and test sets at a 4:1 ratio. All methods were implemented using the default hyperparameters as defined in their respective original papers. For EWC, the regularization term weight was set to 1000 and λ is 15. For LwF, temperature scaling factor T is 2. For WA, ℓ_2 normalization was applied to FC layers. For SCN, β is 0.08. For DER, use 10 warm-up rounds with $\tau = 5$. For CBCNet, the reduction ratio r was set to 32. All methods were trained for 200 epochs with SGD, an initial learning rate of 0.1 (decayed by 0.1 at the 100th and 150th epochs), momentum of 0.9, and weight decay of 2×10^{-4}. The batch size was 128, and the learning rate was reset to 0.1 for each new task. To ensure reproducibility and comparability, the class order is randomized using a numpy random seed before experiments, and the same sequence is used for all methods. Each experiment is run three times with results averaged, and no data augmentation is used. All experiments were performed on an RTX-3090 GPU using Python 3.8 and PyTorch 1.9.

B. Results and Analysis

To comprehensively evaluate CBCNet's performance, we compare it with several state-of-the-art methods. We conducted experiments across three datasets to ensure a fair evaluation and minimize the impact of any single dataset.

The results for the mACC and BWT metrics can be found in Table I. Joint achieved the highest mACC values and represents the ideal scenario, serving as the performance ceiling for CIL methods. Fine-tuning focuses on the current task, leading to severe catastrophic forgetting and the poorest performance among all methods. GEM projects the gradients of new tasks onto a subspace where the output of previous tasks remains unchanged. However, its improvement over fine-tuning is very limited. EWC, as a method based on parameter importance estimation, utilizes Fisher information matrix to constrain changes to critical parameters. However, it achieves only a marginal improvement in mACC compared to fine-tuning. In the context of long-sequence CIL, the above two methods perform as fine-tuning, failing to retain old task knowledge. These methods consider only the effectiveness at the parameter level without utilizing an exemplar set, leading

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to severe overlap in deep feature space. To elaborate, when the classifier is updated solely based on new class data in the context of long-sequence CIL, the feature representations of old classes are not adequately reinforced. This causes the feature representations of old and new classes to become entangled in the deep feature space, reducing the model's ability to discriminate between them. LwF, iCaRL, and WA exhibit a progressive relationship: WA uses postprocessing to correct model biases. Without postprocessing, it becomes iCaRL, which employs data replay and knowledge distillation. iCaRL, when stripped of data replay, reduces to LwF. Finally, removing knowledge distillation, LwF degenerates into fine-tuning. The incremental BWT improvements across these methods highlight the effectiveness of each technique.

SCN, DER, and our method exhibit consistent high performance across all datasets, with relatively stable mACC and BWT values. SCN achieves high mACC across all three datasets, attributed to its asymmetric collaborative dual subnetworks designed for learning new tasks and memorizing old ones. DER employs model duplication and feature concatenation, achieving high mACC but without considering additional storage and computational costs. Our method outperforms the state-of-the-art method, achieving mACC values comparable to those of Joint and the lowest BWT values, which demonstrates the effectiveness of the CBCNet.

The other methods show notable performance variations on the NWPU-45 dataset. This dataset is characterized by substantial variations in translation, viewpoint, orientation, lighting, background, and occlusion, rendering it more complex and diverse than AID and RSI-CB256. For instance, GEM and EWC can effectively limit sudden changes in model parameters when faced with new data from the AID and RSI-CB256 datasets. However, these methods perform as poorly as fine-tuning on NWPU-45 dataset. This suggests that naive regularization approaches may exhibit varying performance depending on the complexity of the dataset. In addition, the larger task number of NWPU-45 dataset may lead to increased task interference. For instance, methods such as iCaRL and WA may perform well on AID and RSI-CB256 datasets but exhibit significantly worse performance on NWPU-45 dataset. Methods that rely heavily on replaying stored samples may lose their effectiveness as the task interference increases.

C. Sequential Learning Analysis

The impact of the number of tasks is a nonnegligible factor for CIL methods. Intuitively, similarity between tasks eases learning, while dissimilarity makes it more challenging. As the number of tasks increases, the model needs greater capacity to differentiate knowledge and the performance tends to decline. Therefore, we analyzed how the models' learning capabilities change as the number of tasks increases in the NWPU-45 dataset.

According to Fig. 5, our method maintains high and relatively stable mACC values across a wide range of task numbers, suggesting that CBCNet effectively adapts to new tasks without compromising previous performance. Fine-tuning starts with a high mACC but shows a rapid decline as the number of tasks increases, indicating catastrophic forgetting. GEM and EWC show a slight mACC advantage over fine-tuning. However, their advantage diminishes with longer task sequences, indicating the limitation of these relatively simple strategies. LwF aims to alleviate catastrophic forgetting through distillation loss and thus outperforms GEM and EWC. iCaRL and WA employ a replay buffer to store exemplars. This straightforward strategy is highly effective from the results. WA addresses the imbalance between new and old classes by adjusting weight averaging, thereby further improving performance. By preserving both long-term and short-term memories, SCN can better handle catastrophic forgetting. Its mACC remains relatively stable and high compared to other methods. DER achieves high mACC across task sequence lengths, which is attributed to its dynamic representation expansion and task-specific networks.

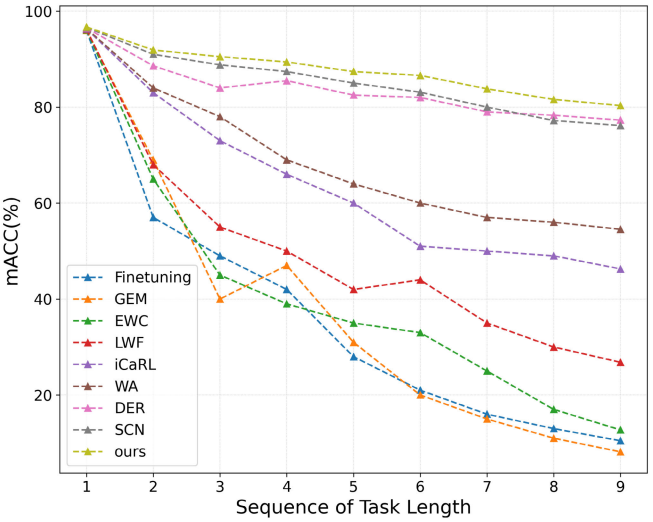


Fig. 5. Comparison of CIL methods on different task sequence length. The NWPU-45 dataset was used for sequential learning analysis. The horizontal axis represents the task sequence length, while the vertical axis indicates mACC values.

D. Optimal Number for Exemplar Selection

In CIL, selecting exemplars is a crucial component of replay-based methods. These exemplars are a small subset of representative samples chosen from previous tasks, which help retain knowledge of old tasks. Too few exemplars can lead to inadequate recall, while too many can consume storage and slowdown training. A moderate number of exemplars balances model performance and resource usage.

Experimental results in Table II show that as the number of exemplars per class increases, both mACC and the training time per epoch rise, reflecting the trade-off between model performance and computational resource consumption. The performance becomes stable with approximately 40 examples per class. Further increases in the number of examples lead to a slowdown in the mACC growth rate, while unnecessarily prolonging training time. Consequently, we select 40 examples per class for all experiments in this article.

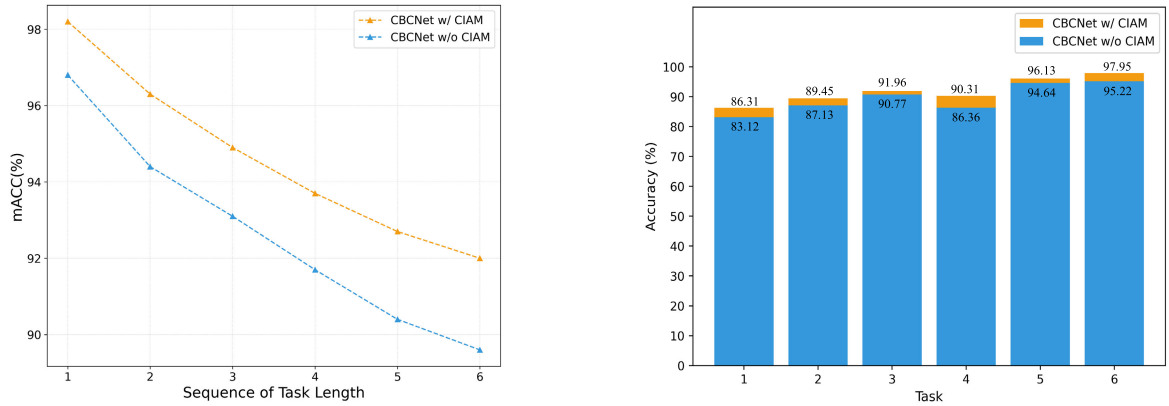
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TABLE II
EFFECTS OF THE NUMBER OF SELECTED EXEMPLARS ON INCREMENTAL LEARNING

Number of exemplars per class	25	30	35	40	45	50
mACC(%)	89.42	89.79	90.24	90.68	90.87	90.98
Training time per epoch(/second)	27.81	34.21	41.06	46.62	53.41	59.81

TABLE III
RESULTS OF ABLATION EXPERIMENTS ON THREE DATASETS FOR CIAM AND CIFM

Method \ Dataset	AID (6 tasks)		RSI-CB256 (7 tasks)		NWPU-45 (9 tasks)	
	mACC(%)	BWT(%) ↓	mACC(%)	BWT(%)↓	mACC(%)	BWT(%)↓
w/o CIFM	88.45	9.07	89.24	13.91	79.09	20.73
w/o CIAM	87.94	11.04	88.31	14.17	77.72	24.73
w/o CIAM and CIFM	86.91	15.92	86.75	17.24	75.49	30.91
CBCNet	**89.27**	**9.29**	**90.68**	**10.17**	**80.31**	**19.61**



(a) (b)

Fig. 6. Ablation experiments for CIAM. The six-task PatternNet dataset was used for sequential learning analysis. (a) mACC and (b) accuracy comparisons including and excluding CIAM.

E. Ablation Experiments

To further illustrate the contribution, we conduct thorough ablation experiments focusing on the key components in CBCNet, including CIAM, CIFM, TCDM, and MKDM.

1) Effect of CIAM: We conducted ablation studies of the CIAM on three datasets, and the results are presented in Table III. When compared with the network without CIAM (w/o CIAM), CBCNet achieves a higher mACC and a lower BWT value, indicating that CIAM effectively enhances feature representations and alleviates the issue of forgetting.

To determine if its contribution is justified, we further performed sequential learning analysis on the six-task PatternNet dataset. Fig. 6(a) illustrates that the mACC values of the model equipped with CIAM are consistently higher throughout the incremental learning phases. In Fig. 6(b), the accuracy on each task after learning all tasks is displayed. On the six tasks, the accuracy is notably higher by 3.19%, 2.32%, 1.19%, 3.95%, 1.49%, and 2.73%, respectively. The results are attributed to the fact that CIAM highlights informative regions and suppresses unnecessary ones within RSIs.

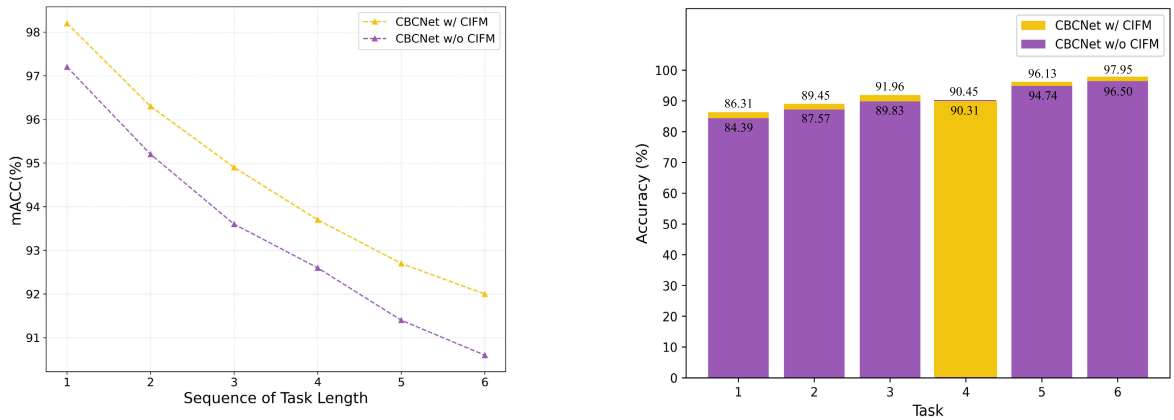
2) Effect of CIFM: To verify CIFM's effectiveness, we conducted experiments across three datasets, and the results are summarized in Table III. The CBCNet attains a higher mACC and a lower BWT value than the network lacking CIFM. We further performed sequential learning analysis on the six-task PatternNet dataset. Fig. 7(a) and (b) illustrates the mACC and accuracy for methods with and without CIFM, respectively. When using CIFM, there is only a slight decrease in accuracy on the fourth task. Notably, the model with CIFM shows significant improvements in accuracy by 1.92%, 1.88%, 2.13%, 1.39%, and 1.45% on the other tasks, respectively.

The situation on the fourth task may arise from two reasons: varying sensitivities to multilevel features across different RSIs, and model capacity limitations that improve performance in other stages but compromise accuracy in the current one. However, the performance gains on other tasks boost our confidence in CIFM.

3) Joint Effect of CIFM and CIAM: In our model, CIAM and CIFM operate synergistically to enhance the quality of feature representations. Therefore, we present a comprehensive evaluation of the joint effect of CIFM and CIAM.

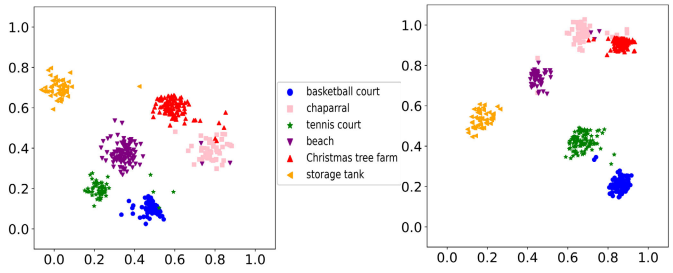
We conducted ablation experiments for CIAM and CIFM on three datasets, and the results are presented in Table III. The experimental results show that the mACC increases by 2.36%, 3.93%, and 4.82%, and BWT decreases by 6.63%, 7.07%, and 11.3% on the three datasets when using the two

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(a) (b)

Fig. 7. Ablation experiments for CIFM. The six-task PatternNet dataset was used for sequential learning analysis. (a) mACC and (b) accuracy comparisons including and excluding CIFM.



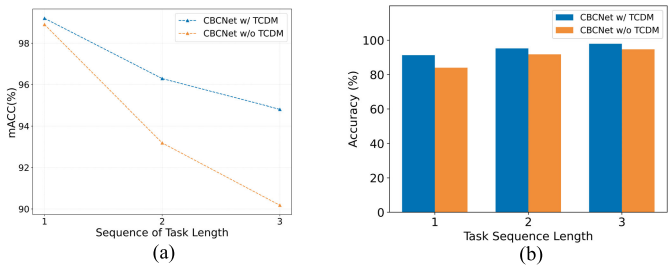
(a) (b)

Fig. 8. Ablation experiments for the joint effect of CIFM and CIAM on six-task PatternNet dataset. Feature embeddings for the six classes in the first task were visualized using t-SNE. Each point represents a sample, with colors and shapes employed to differentiate between distinct classes. The legend, composed of markers that represent different classes, is displayed in the center. (a) CBCNet without CIAM and CIFM. (b) CBCNet with CIAM and CIFM.

modules simultaneously in our method. These improvements are significantly greater than those achieved when each module is utilized independently, indicating a synergistic effect between the two components.

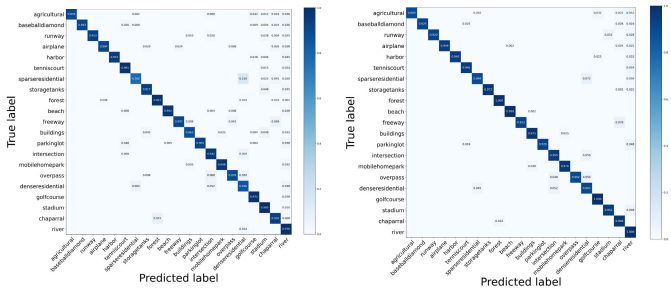
T-distributed stochastic neighbor embedding (t-SNE) [70] is a nonlinear dimensionality reduction technique for visualizing high-dimensional data. To further substantiate whether CIFM and CIAM sufficiently learn robust feature representations and capture effective class discrimination, t-SNE is used to map the final layer's high-dimensional features of CBCNet to 2-D for visualization.

After completing all learning phases on the three-task UC-Merced dataset, the feature embeddings of the six classes in the first task were visualized using t-SNE. As illustrated in Fig. 8(b), the t-SNE dimensionality-reduced data shows a clearer clustering structure. Samples of the same class are more tightly clustered together in the feature space, causing tighter intraclass aggregation and amplifying the feature differences between different classes, which increases the interclass distance. This is because CIAM and CIFM collaborate to highlight informative regions across multiple scales and aggregate contextual information from multiple levels within RSIs, constructing a more distinguishable feature representation.



(a) (b)

Fig. 9. Ablation experiments for TCDM on the three-split UC-Merced dataset. (a) mACC and (b) accuracy comparisons including and excluding TCDM. CBCNet without TCDM refers to the model using the standard multiclass CE loss.



(a) (b)

Fig. 10. Comparison of confusion matrices on the UC-Merced dataset. (a) CBCNet without TCDM. (b) CBCNet with TCDM.

4) Effect of TCDM: To illustrate the contribution of TCDM, we conducted ablation experiments on TCDM across the three-split UC-Merced dataset, and results are shown in Fig. 9. Fig. 9(a) shows that the mACC values of the model with TCDM are consistently higher at each incremental learning phase. Fig. 9(b) shows the accuracy on each task after learning all the tasks, and the model with TCDM improves accuracy by 7.28%, 3.42%, and 3.21% on the respective tasks. This improvement is due to the module's ability to reduce the disproportionate impact on the probabilities of the old and new classes, thereby achieving higher classification accuracy.

To visualize the preferences of the methods during the prediction phase, we compared the confusion matrices for the models with and without TCDM on the UC-Merced dataset. Compared to Fig. 10(a), Fig. (b) shows a notable increase

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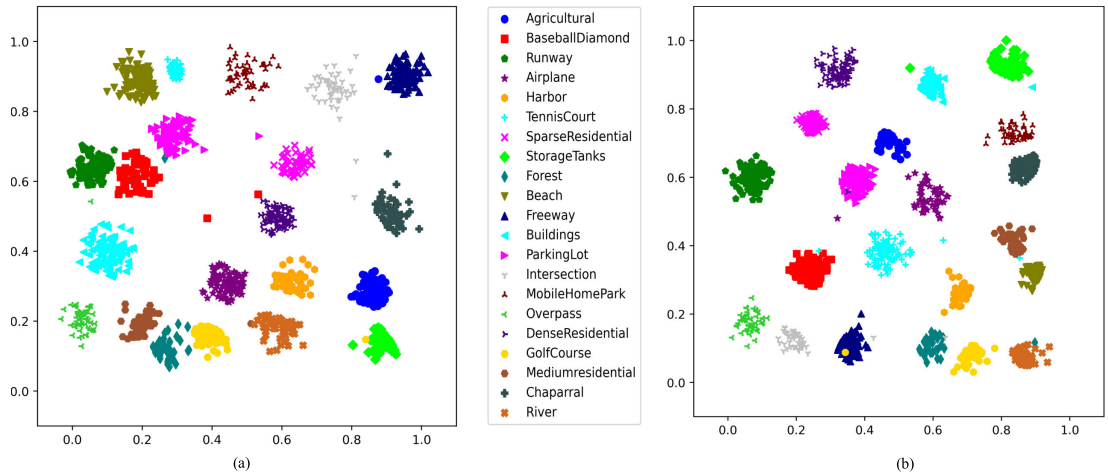


Fig. 11. Ablation experiments for TCDM on the UC-Merced dataset. Feature embeddings for the 21 classes were visualized using t-SNE. Each point represents a sample, while colors and shapes are utilized to distinguish between different classes. The legend, composed of markers that represent different classes, is displayed in the center. (a) CBCNet without TCDM. (b) CBCNet with TCDM.

TABLE IV
RESULTS OF ABLATION EXPERIMENTS ON THREE DATASETS FOR MKDM

$\mathcal{L}$			AID(6 tasks)		RSI-CB256(7 tasks)		NWPU-45(9 tasks)	
$\mathcal{L}_{\text{GB-CE}}$	$\mathcal{L}_{\text{TWD}}$	$\mathcal{L}_{\text{HPD}}$	mACC(%)	BWT(%)↓	mACC(%)	BWT(%)↓	mACC(%)	BWT(%)↓
✓			86.42	13.37	87.29	14.46	76.49	25.04
✓	✓		88.17	10.82	89.93	9.95	79.21	22.96
✓		✓	87.69	11.95	89.47	10.67	79.04	23.17
✓	✓	✓	**89.27**	**9.29**	**90.68**	**10.17**	**80.31**	**19.61**

in the diagonal elements, particularly for previously existing classes. Simultaneously, off-diagonal elements decrease and the entire confusion matrix is more balanced. These improvements can be attributed to TCDM, which avoids mutual interference between new and old classes by independently computing their output scores. This approach effectively alleviates classification bias issues arising from data imbalance.

The feature embeddings of the UC-Merced dataset are visualized using t-SNE. Fig. 11(b) shows a more regular and ordered data distribution compared to the less organized and diffuse pattern in Fig. 11(a), which lacks clear boundaries. These observations suggest that CBCNet with TCDM has effectively achieved data clustering. The t-SNE plot for CBCNet with TCDM shows larger interclass distances, reducing interclass correlation and enhancing classification feasibility. Furthermore, high degree of intraclass compactness in Fig. 11(b) signifies superior classification performance. In summary, CBCNet with TCDM outperforms in terms of classification accuracy, interclass distance, and intraclass aggregation.

5) Effect of MKDM: Our method integrates two knowledge distillation losses within MKDM. $\mathcal{L}_{\text{TWD}}$ transfers knowledge in output probability space, whereas $\mathcal{L}_{\text{HDP}}$ does so in feature representation space. To validate the effectiveness of this dual-constraint knowledge distillation module, we conducted ablation experiments on three datasets, as detailed in Table IV.

Compared to models trained solely with classification loss, adding distillation loss significantly improved mACC and reduced BWT. This demonstrates the effectiveness of distillation losses in guiding model training in both the output probability space and the feature representation space. Specifically, using $\mathcal{L}_{\text{TWD}}$ alone resulted in slightly higher performance than using $\mathcal{L}_{\text{HDP}}$ alone, suggesting that constraints in the output probability space will be of higher priority. The experimental results show that mACC increased by 2.85%, 3.39%, and 3.82%, and BWT decreased by 4.08%, 4.29%, and 5.43% on the three datasets when both distillation losses are used simultaneously in our method. These outcomes exceed the performance achieved by using distillation loss alone, indicating a synergistic effect from dual distillation at both the feature and semantic levels.

To visualize prediction preferences, we compared confusion matrices for models with and without MKDM on the UC-Merced dataset. As shown in Fig. 12(b), the confusion matrix exhibits a reduction in off-diagonal elements and a more uniform diagonal, indicating a balanced predictive performance across all classes. The diagonal elements are brighter for all classes in CBCNet with MKDM. After incorporating knowledge distillation strategy, the model better balances predictions between new and old classes, leading to more accurate predictions for the old classes.

F. Effect of Hyperparameters

The reduction ratio r in the second-stage attention mechanism serves as a hyperparameter that controls channelwise compression, reducing computational load while preserving feature capture. The selection of r involves a trade-off between model performance and efficiency. A larger r enhances model efficiency by reducing channels but may lose fine-grained

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TABLE V
EFFECTS OF THE REDUCTION RATIO ON MODEL PERFORMANCE AND EFFICIENCY

The reduction ratio r	1	4	8	16	32	64
mACC(%)	90.83	90.76	90.72	90.70	90.68	90.34
Training time per epoch(/second)	79.39	68.27	61.07	56.89	46.62	37.75

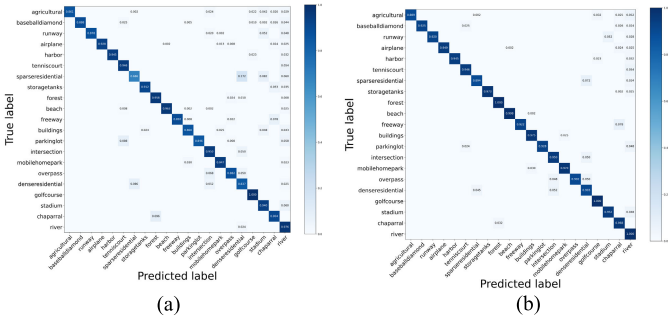


Fig. 12. Comparison of confusion matrices on the UC-Merced dataset. (a) CBCNet without MKDM. (b) CBCNet with MKDM.

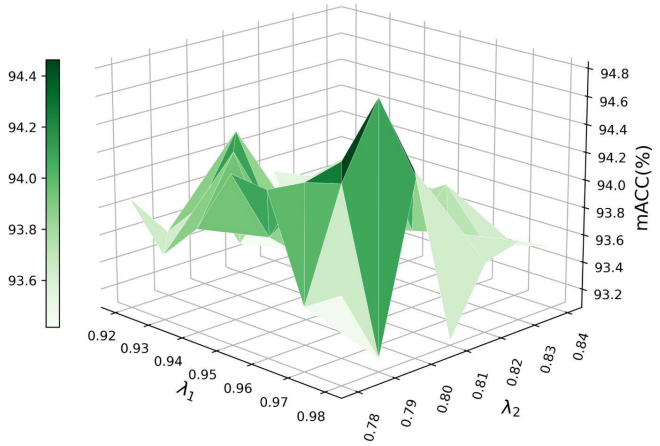


Fig. 13. mACC versus the hyperparameter values of λ_1 and λ_1 on the three-task UC-Merced dataset.

details, whereas a smaller r retains more details at the expense of increased computational cost.

As shown in Table V, we performed ablation studies on the RSI-CB256 dataset. As the reduction ratio r increases, both mACC and the training time per epoch decrease, reflecting the trade-off between model performance and efficiency. When the reduction ratio r reaches 64, the mACC experiences an abrupt decline, which is likely attributable to the model overfitting due to an excessively reduced number of channels. In particular, we found that setting $r = 32$ achieved a good balance between model performance and efficiency. Consequently, we adopted this value for all experiments.

Our approach also incorporates two other key hyperparameters: λ_1 and λ_2. As illustrated in (28), these hyperparameters serve as balance factors in the total loss. λ_1 modulates the influence of knowledge distillation in the output probability space, whereas λ_2 regulates the impact of knowledge distillation within the feature representation space. When the balance factors are small, the model experiences severe catastrophic forgetting due to inadequate constraint. Conversely, as these values increase, the model places greater emphasis on the accuracy of knowledge transfer and retention of prior knowledge, resulting in stricter loss, more rigid constraints, and lower plasticity.

To further illustrate the effects of these two key hyperparameters in combating catastrophic forgetting, we performed ablation experiments on the values of the two balancing factors using the three-task UC-Merced dataset. Initially, we observed that the model's performance was impressive when λ_1 ranged from 0.92 to 0.98 and λ_2 ranged from 0.78 to 0.84. Consequently, we executed comprehensive experiments within this range, incrementing λ_1 and λ_2 by 0.01 at each step.

As depicted in Fig. 13, the model attains its fullest capacity when λ_1 is set to 0.97 and λ_2 to 0.80 for the UC-Merced dataset. Consequently, in the experiments conducted for this study, we utilized these values for λ_1 and λ_2.

V. CONCLUSION

In this article, we propose an effective CIL framework for RCCS, aiming to effectively address catastrophic forgetting and balance the stability–plasticity dilemma. It mainly consists of four novel components, i.e., CIAM, CIFM, TCDM, and MKDM. Specifically, CIAM is designed to capture cross-dimensional attention maps that effectively incorporate channel, position, and direction-aware information, which are then applied to the input feature to construct a richer feature representation. Meanwhile, CIFM is proposed to explore the correlations among multilevel features and capture rich contextual information through their fusion. These two novel components collaborate synergistically to attain a holistic understanding of the RS scenes. Besides, we conduct a systematic analysis to identify the root causes of prediction bias toward new classes in CIL methods and then propose a novel TCDM. It imposes a constraint to mitigate prediction bias toward new classes and optimizes the model's classification capabilities across all seen classes. Meanwhile, an adaptive gradient-balanced CE loss is introduced to adjust for the skewed gradient contributions to the classification scores. In addition, most methods typically focus solely on constraining the output probabilities for past tasks. To counteract this constraint, MKDM is designed to empower the model to maintain critical representations in the feature space and make well-informed decisions in the output probability space.

To demonstrate the outperformance and robustness of our proposed CBCNet, extensive experiments and thorough analysis are conducted. Three datasets were utilized for algorithm comparisons with the selected state-of-the-art methods, and five datasets with complex CIL scenarios were employed in ablation studies to verify the performance of novel components

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proposed in our framework. In future work, we will explore more efficient modules to address the computational and memory overhead inherent in our method. Furthermore, we will extend the proposed framework to few-shot CIL. It requires the model to access only a limited amount of new data when learning new classes, while avoiding forgetting previously acquired knowledge. This setup more closely aligns with real-world application scenarios.

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